

$$\mathbf{x}_t = (f_{1,t}, f_{2,t}, \dots, f_{p,t})^\top \in \mathbb{R}^p$$

$$\boldsymbol{\mu}_t = (\mu_{1,t}, \dots, \mu_{p,t})^\top$$

$$\mathbf{r}_t = \mathbf{x}_t - \boldsymbol{\mu}_t$$

$$\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$$

$$\begin{aligned} d_t^2 &= \mathbf{r}_t^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}_t \\ \text{\quad} \\ d_t &= \sqrt{d_t^2} \end{aligned}$$